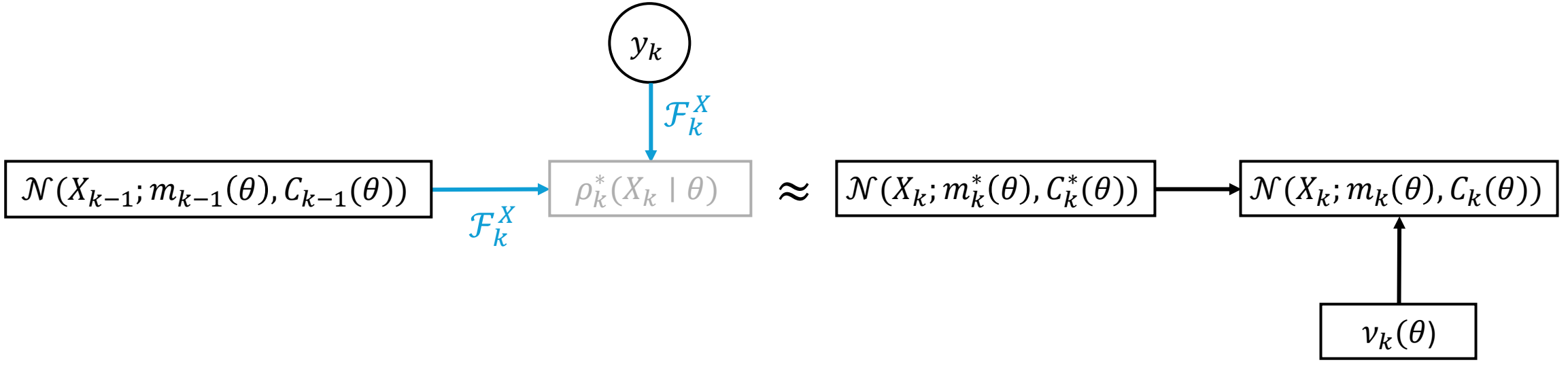
y_k
$\mathcal{F}_k^X$
$\mathcal{N}(X_{k-1}; m_{k-1}(\theta), C_{k-1}(\theta))$
$\mathcal{F}_k^X$
$\rho_k^*(X_k \mid \theta)$
$\approx$
$\mathcal{N}(X_k; m_k^*(\theta), C_k^*(\theta))$
$\mathcal{N}(X_k; m_k(\theta), C_k(\theta))$
$\nu_k(\theta)$